TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	88.4 ± 0.66
Correlation Stability	89.3 ± 1.35
Distribution Consistency	88.8 ± 1.08
Entropy Preservation	85.2 ± 0.60
Mean Z Score Drift	0.1 ± 0.03
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.2 ± 0.40
Data Density	100.0 ± 0.00
Coverage Score	96.9 ± 1.64
Feature Accuracy	92.8 ± 1.54
Novelty Score	0.8 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: breast_cancer_coimbra.csv

1. Dataset Portrait

Attribute	Value
Rows	116
Columns	10
Numeric features	10
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	57.302	57.457	+0.155	0.3%	OK
BMI	27.582	27.957	+0.375	1.4%	OK
Glucose	97.793	99.120	+1.326	1.4%	OK
Insulin	10.012	11.134	+1.122	11.2%	Warn
HOMA	2.695	2.937	+0.242	9.0%	OK
Leptin	26.615	25.449	-1.166	4.4%	OK
Adiponectin	10.181	9.045	-1.136	11.2%	Warn
Resistin	14.726	16.646	+1.920	13.0%	Warn
MCP.1	534.647	548.886	+14.239	2.7%	OK
Classification	0.552	0.500	-0.052	9.4%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	88.4 ± 0.66
Correlation Stability (*2)	89.3 ± 1.35
Distribution Consistency (*3)	88.8 ± 1.08
Entropy Preservation (*4)	85.2 ± 0.60
Mean Z Score Drift (*5)	0.1 ± 0.03
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.2 ± 0.40
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	96.9 ± 1.64

Feature Accuracy (*10)	92.8 ± 1.54
Novelty Score (*11)	0.8 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4958333333333333	0.5125	0.4541666666666667	-0.04166666666666665	
F1	0.6547289527475286	0.567640800669385	0.5783285081853193	-0.07640044456220926	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7458333333333333	0.6208333333333333	0.7	-0.04583333333333332	
F1	0.754839950419102	0.6213151119567669	0.707844142992472	-0.046995807426629954	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7041666666666667	0.6833333333333333	0.7375	0.03333333333333336	
F1	0.7207707755207754	0.7053260898151354	0.7508240102500392	0.030053234729263655	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4625	0.5499999999999999	0.4791666666666667	0.01666666666666663	
F1	0.5361601301923884	0.5699557632766092	0.5350343156498901	-0.0011258145424980549	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5875	0.5875000000000001	0.5666666666666668	-0.020833333333333332	
F1	0.5956336213182747	0.6249295532054153	0.6491719293433127	0.05353830802503784	

Model: DT

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7416666666666667	0.5666666666666667	0.7333333333333333	-0.00833333333333337	
F1	0.7509975676497416	0.5557509157509157	0.7326373256373256	-0.01836024201241594	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7541666666666667	0.6458333333333333	0.7458333333333333	-0.00833333333333336	
F1	0.774786398467433	0.6631620553359685	0.7528388332098009	-0.02194756525763194	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6416666666666667	0.5916666666666667	0.6499999999999999	0.00833333333333337	
F1	0.5557236965932618	0.5740525327187731	0.5682397504456327	0.012516053852370996	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7083333333333334	0.6291666666666667	0.6416666666666667	-0.06666666666666668	
F1	0.7179811943844927	0.6495913715913716	0.6608170302683047	-0.05716416411618809	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.725	0.6916666666666667	0.7041666666666666	-0.02083333333333336	
F1	0.7359528619528619	0.7065644969722931	0.7145314456464527	-0.02142141630640929	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	48.0	23.5	70.0	2.707	0.467408667	8.8071	9.7024	7.99585	417.114	0.0
2	83.0	20.69049454	92.0	3.115	0.706897333	8.8438	5.429285	4.06405	468.786	0.0
3	82.0	23.12467037	91.0	4.498	1.009651067	17.9393	22.43204	9.27715	554.697	0.0
4	68.0	21.36752137	77.0	3.226	0.612724933	9.8827	7.16956	12.766	928.22	0.0
5	86.0	21.11111111	92.0	3.549	0.8053864	6.6994	4.81924	10.57635	773.92	0.0
6	49.0	22.85445769	92.0	3.226	0.732086933	6.8317	13.67975	10.3176	530.41	0.0
7	89.0	22.7	77.0	4.69	0.890787333	6.964	5.589865	12.9361	1256.083	0.0
8	76.0	23.8	118.0	6.47	1.883201333	4.311	13.25132	5.1042	280.694	0.0
9	73.0	22.0	97.0	3.35	0.801543333	4.47	10.358725	6.28445	136.855	0.0
10	75.0	23.0	83.0	4.952	1.013839467	17.127	11.57899	7.0913	318.302	0.0
11	34.0	21.47	78.0	3.469	0.6674356	14.57	13.11	6.92	354.6	0.0
12	29.0	23.01	82.0	5.663	1.145436133	35.59	26.72	4.58	174.8	0.0
13	25.0	22.86	82.0	4.09	0.827270667	20.45	23.67	5.14	313.73	0.0
14	24.0	18.67	88.0	6.107	1.33	8.88	36.06	6.85	632.22	0.0
15	38.0	23.34	75.0	5.782	1.06967	15.26	17.95	9.35	165.02	0.0
16	44.0	20.76	86.0	7.553	1.6	14.09	20.32	7.64	63.61	0.0
17	47.0	22.03	84.0	2.869	0.59	26.65	38.04	3.32	191.72	0.0
18	61.0	32.03895937	85.0	18.077	3.790144333	30.7729	7.780255	13.68392	444.395	0.0
19	64.0	34.5297228	95.0	4.427	1.037393667	21.2117	5.46262	6.70188	252.449	0.0
20	32.0	36.51263743	87.0	14.026	3.0099796	49.3727	5.1	17.10223	588.46	0.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	89.0	26.66666667	77.0	3.855	0.605507467	14.3224	9.7024	7.99585	698.789	0.0
2	71.0	21.51385851	88.0	6.107	1.245642	8.88	5.589865	5.57055	483.377	1.0
3	44.0	27.88761707	101.0	22.033	6.4834952	22.8884	21.57	7.32	136.855	1.0
4	67.0	30.3	90.0	5.537	1.133929133	12.331	12.1	16.43706	910.489	0.0
5	74.0	25.59	112.0	26.211	6.4834952	19.0653	5.4861	19.46324	552.444	1.0
6	71.0	32.5	102.0	5.43	1.4066068	47.647	21.42	21.44366	270.142	0.0
7	64.0	22.22222222	98.0	5.7	1.37788	12.1905	4.783985	13.91245	395.976	1.0
8	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0
9	68.0	21.08281329	102.0	6.2	1.55992	9.6994	8.574655	13.74244	448.799	1.0
10	42.0	21.35991456	93.0	2.999	0.6879706	19.0826	8.462915	17.37615	321.919	1.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	52.0	30.8012487	85.0	30.212	9.736007333	31.0385	7.780255	49.24184	1102.11	1.0
2	45.0	23.14049587	86.0	4.713	1.14478	17.9973	8.412175	31.6904	407.206	1.0
3	48.0	28.125	94.0	18.2	3.790144333	21.778	2.19428	19.46324	910.489	1.0
4	60.0	29.15451895	100.0	8.079	2.628282667	14.7485	10.26266	23.1177	621.273	1.0
5	54.0	31.23140988	119.0	10.395	2.534931667	56.502	9.92365	5.31	198.4	1.0
6	47.0	28.125	75.0	3.855	0.827270667	14.9084	3.19209	6.92	392.46	0.0
7	50.0	38.57875854	106.0	6.703	1.752611067	46.6401	4.667645	11.78388	887.16	0.0
8	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0
9	38.0	22.4996371	95.0	5.261	1.232827667	8.438	4.77192	15.73606	199.055	1.0
10	81.0	31.64036818	100.0	9.669	2.38502	38.8066	10.636525	29.5583	426.175	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	44.0	27.88761707	93.0	8.396	1.8404096	12.6757	8.42996	18.35574	407.206	1.0
2	66.0	26.34929208	77.0	5.819	1.1006464	17.9393	8.6	11.78796	928.22	0.0
3	69.0	25.59	86.0	2.82	0.6538048	65.926	7.16956	10.34455	910.489	1.0
4	42.0	21.35991456	106.0	12.162	4.468268	4.47	21.056625	7.84	90.09	1.0
5	69.0	21.36752137	98.0	6.03	1.883201333	32.58	4.230105	10.57635	713.239	1.0
6	61.0	34.5297228	101.0	5.81	1.370998	18.1314	11.78796	9.1539	585.307	0.0
7	43.0	31.25	103.0	4.328	1.099600533	25.7816	12.71896	38.6531	775.322	1.0
8	24.0	18.67	88.0	6.107	1.33	8.88	36.06	6.85	632.22	0.0
9	71.0	25.51020408	112.0	10.395	2.871792	19.0653	5.4861	42.7447	799.898	1.0
10	49.0	21.36752137	78.0	2.64	0.507936	6.3339	3.886145	22.94254	737.672	1.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	57.0	34.83814777	134.0	19.91	3.869788067	37.843	20.03	4.62	572.783	1.0
2	72.0	21.51385851	112.0	4.902	1.4026256	32.58	7.16956	5.57055	355.31	1.0
3	85.0	26.6	138.0	24.887	7.111918	65.926	5.357135	49.24184	1256.083	0.0
4	66.0	31.44654088	89.0	15.533	3.790144333	45.272	8.2863	6.71026	960.246	0.0
5	65.0	26.5625	89.0	6.524	1.55992	14.9084	6.644245	18.35574	713.239	1.0
6	86.0	27.18	92.0	3.115	0.617890133	28.562	6.644245	13.91245	1698.44	1.0
7	75.0	30.48	152.0	7.01	2.628282667	50.53	10.06	11.73	99.45	1.0
8	60.0	26.34929208	103.0	5.138	1.305394533	24.2998	2.19428	20.2535	378.996	0.0
9	77.0	35.58792924	76.0	3.881	0.727558133	21.7863	8.12555	17.2615	618.272	0.0
10	72.0	23.62	105.0	4.42	1.14478	21.78	17.86	4.82	195.94	1.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	36.0	34.17489	80.0	4.53	0.890787333	14.3224	11.78796	5.62592	209.19	0.0
2	61.0	32.03895937	93.0	10.491	1.6	26.8081	8.42996	10.19299	473.859	0.0
3	54.0	22.83287935	86.0	3.855	0.791257333	24.2998	4.230105	18.35574	655.834	1.0
4	45.0	29.38475666	90.0	4.376	1.14478	18.69	12.1	14.91922	269.487	1.0
5	78.0	21.11111111	97.0	6.862	1.300426667	9.62	22.43204	8.4156	330.16	0.0
6	82.0	25.3	94.0	12.305	3.0099796	4.47	13.25132	8.4156	220.66	0.0
7	77.0	35.58792924	76.0	3.881	0.727558133	21.7863	8.12555	17.2615	618.272	0.0
8	48.0	28.125	90.0	2.54	0.56388	15.5325	10.22231	16.11032	1698.44	1.0
9	58.0	29.15451895	139.0	16.582	5.685415067	22.8884	10.26266	13.97399	923.886	1.0
10	68.0	21.36752137	77.0	3.226	0.612724933	9.8827	7.16956	12.766	928.22	0.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	44.0	22.03	84.0	5.782	1.245642	17.9393	17.95	5.14	468.786	0.0
2	86.0	20.69049454	92.0	4.364	1.0011016	9.62	8.574655	4.06405	313.73	0.0
3	86.0	26.6	138.0	30.13	9.736007333	65.926	21.42	5.2633	215.769	1.0
4	57.0	36.05	131.0	8.15	2.628282667	17.87	6.644245	4.19	244.75	1.0
5	75.0	23.62	138.0	51.814	25.05034187	4.311	22.43204	3.21	280.694	1.0
6	75.0	25.9	85.0	4.58	0.832352	24.2998	3.19209	7.5767	518.586	0.0
7	74.0	28.65013774	88.0	3.012	0.6538048	31.1233	7.65222	18.35574	572.401	1.0
8	65.0	32.05	97.0	5.73	1.370998	61.48	22.54	10.33	314.05	1.0
9	76.0	27.2	94.0	14.07	3.262364	35.891	9.34663	8.4156	377.227	0.0
10	65.0	29.666548	85.0	14.649	3.071407	26.5166	7.28287	19.46324	1698.44	1.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	89.0	26.6	92.0	4.498	1.14478	7.85	9.000805	9.1539	318.302	0.0
2	47.0	38.57875854	75.0	3.188	0.612724933	24.96	4.617125	23.3819	695.754	0.0
3	45.0	26.85	94.0	2.432	0.605507467	54.68	22.54	7.32	632.22	1.0
4	48.0	20.82999519	94.0	4.462	1.099600533	11.2406	16.1	5.31	63.61	1.0
5	59.0	22.65625	93.0	3.226	0.59	9.8648	3.886145	13.74244	713.239	1.0
6	68.0	21.51385851	104.0	6.03	1.341324	14.9037	7.16956	8.2049	225.88	1.0
7	35.0	30.27681661	84.0	4.376	0.9067072	39.2134	9.048185	16.43706	733.797	0.0
8	69.0	32.5	93.0	5.43	1.245642	15.145	11.78796	11.78796	270.142	0.0
9	58.0	29.15451895	139.0	16.582	5.685415067	22.8884	10.26266	13.97399	923.886	1.0
10	49.0	29.77777778	70.0	8.396	1.449709333	51.3387	10.73174	20.76801	602.486	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	43.0	34.42217362	87.0	14.026	4.468268	49.3727	6.966895	17.10223	1698.44	0.0
2	46.0	23.14049587	92.0	3.549	0.790181867	8.6874	5.589865	8.2049	45.843	1.0
3	76.0	25.9	94.0	4.462	1.037393667	10.39	21.42	7.5767	417.114	0.0
4	62.0	26.84	100.0	5.73	1.37788	83.4821	5.47817	82.1	426.175	1.0
5	72.0	37.109375	97.0	18.077	2.940414667	31.2128	20.03	10.15726	269.487	1.0
6	86.0	26.6	100.0	6.2	1.432235467	61.48	5.357135	10.33	232.018	1.0
7	45.0	37.03560819	83.0	6.76	1.383997333	39.9802	4.617125	8.70448	586.173	0.0
8	61.0	32.03895937	85.0	18.077	3.790144333	30.7729	7.780255	13.68392	444.395	0.0
9	46.0	22.21	86.0	36.94	7.836205333	10.16	9.76	5.68	312.0	1.0
10	85.0	27.68877813	196.0	51.814	25.05034187	70.8824	7.901685	55.2153	1078.359	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	72.0	31.64036818	139.0	5.636	1.752611067	47.647	4.104105	20.4685	1698.44	1.0
2	66.0	36.21227888	101.0	9.669	2.5101466	17.87	6.420295	8.70448	209.19	0.0
3	89.0	20.69049454	92.0	3.42	0.732193	9.8827	11.57899	9.27715	698.789	0.0
4	46.0	28.125	116.0	58.46	7.836205333	46.6401	1.65602	17.55503	407.206	1.0
5	89.0	22.7	97.0	6.862	1.55992	17.127	4.294705	11.78388	530.41	0.0
6	58.0	36.05	131.0	8.15	2.633536667	74.7069	6.796985	15.55625	638.261	1.0
7	78.0	25.3	60.0	3.508	0.519184	6.633	10.567295	4.6638	209.749	0.0
8	52.0	30.8012487	87.0	30.212	6.4834952	29.2739	6.26854	24.24591	764.667	1.0
9	45.0	20.26	92.0	3.44	0.780650667	7.65	16.67	7.84	193.87	1.0
10	59.0	28.67262608	77.0	3.188	0.605507467	17.022	16.44048	31.6904	910.489	1.0

TabuLLM: AI Benchmark Report

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	BMI	Glucose	Insulin	HOMA	Leptin	Adiponectin	Resistin	MCP.1	Classification
1	76.0	23.8	103.0	4.53	1.008511467	17.127	11.57899	4.6638	200.976	0.0
2	36.0	34.17489	86.0	10.175	2.534931667	56.502	22.54	4.35	483.377	0.0
3	65.0	32.05	119.0	15.89	4.468268	46.076	21.57	4.62	244.75	1.0
4	48.0	22.21	74.0	4.364	0.890787333	9.8648	8.237405	8.89	209.749	1.0
5	45.0	22.85445769	70.0	10.555	2.3464512	35.59	17.95	10.15726	635.049	0.0
6	44.0	22.85445769	84.0	7.553	1.449709333	18.16	13.11	6.92	191.72	0.0
7	64.0	34.5297228	95.0	4.427	1.037393667	21.2117	5.46262	6.70188	252.449	0.0
8	32.0	36.51263743	87.0	14.026	3.0099796	49.3727	5.1	17.10223	588.46	0.0
9	43.0	31.25	103.0	4.328	1.099600533	25.7816	12.71896	38.6531	775.322	1.0
10	57.0	34.83814777	95.0	12.548	2.940414667	33.1612	2.36495	9.9542	655.834	1.0